Math 340 **Name (Print):** ______________________
Practice Exam 2 **TA's name and Section#:** ______________________
Time Limit: 75 Minutes **Student ID Number:** ______________________

This exam contains 6 pages (including this cover page) and 8 problems. The last page is the scratch paper. If any pages are missing, please inform the instructor immediately.

Directions:

- Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.
- You may *not* use your calculators, books and notes.
- You must show all your work. A stranger should be able to pick up your paper and follow your reasoning. You must use proper notation and define the events clearly. Partial credits may be given.
- Unless otherwise noted, your answers must be in fraction form or terminating decimals.
- Circle your answer to the True/False problems and justify your answer.
- For the free response problems you need to show all necessary work in the space provided for each problem. Answers must be justified with sufficient work, and partial credit will be given for appropriate work shown.

I do not lie, cheat or steal, or tolerate those who do.

By signing below you indicate that all your work is your own and that you have neither given nor received help from external sources.

Signature: ______________________

T/F	
P.6	
P.7	
P.8	
Total	

True/False Questions: If true, give your justification. If false, give a counterexample or some reasoning.

1. (2 points) Let W be the set of all vectors of the form $\begin{bmatrix} x \\ y \end{bmatrix}$, where $x + y = 1$. W is a subspace of $\mathbb{R}^2$ with the usual operations of vector addition and scalar multiplication.
   A. True.
   B. False.

2. (2 points) Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be a spanning set of the vector space $\mathbb{R}^3$. If $\mathbf{v}_1$ is a linear combination of $\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$, then $\{\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is also a spanning set of $\mathbb{R}^3$.
   A. True.
   B. False.

3. (2 points) If a linear transformation $L : \mathbb{R}^3 \to \mathbb{R}^3$ is one-to-one, then L is invertible.
   A. True.
   B. False.

4. (2 points) Let $L : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $L([u_1\ u_2\ u_3]^T) = [0\ \ u_1^2\ \ u_2 + u_3]^T$. Then L is a linear transformation.
   A. True.
   B. False.

5. (2 points) Let S and T be two finite nonempty subsets of a vector space such that S is contained in T. If T is linearly *dependent*, so is S.
   A. True.
   B. False.

6. (10 points) Let $L : \mathbb{R}^5 \to \mathbb{R}^4$ be defined by

$$L\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 & -1 & 3 & -1 \\ 1 & 1 & 0 & 2 & -1 \\ 2 & 1 & -1 & 5 & -1 \\ 0 & -1 & -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix}.$$

(a) Find a basis for $\mathrm{Ker}(L)$.

(b) What is the dimension of $\mathrm{Ker}(L)$?

(c) Find a basis for $\mathrm{range}(L)$ (that is, the same thing as $\mathrm{im}(L)$).

(d) What is the dimension of $\mathrm{range}(L)$?

7. (10 points) Let W be the subspace of $\mathbb{R}^4$ spanned by the set of vectors $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -6 \\ -3 \end{bmatrix}, \begin{bmatrix} -1 \\ -5 \\ 1 \\ 0 \end{bmatrix} \right\}$.

   Find the basis for W. What is the dimension of W?

8. (10 points) Let P_2 denote the vector space of polynomials in t of degree at most 2. For what values of λ are the vectors $\mathbf{v}_1 = 2t^2+t+2$, $\mathbf{v}_2 = t^2+\lambda$ and $\mathbf{v}_3 = t^2+2t+\lambda^2$ linearly independent?

Scratch Paper